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(54) Title: **Uav-based system for detection and geolocation of underground pipeline structural deformation**

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(57) Abstract:

A UAV-based system for detecting structural deformation in underground pipelines is disclosed. The system comprises a thermal imaging module configured to capture subsurface thermal variations, a preprocessing module configured to convert thermal data into color-mapped images, and an onboard edge computing module configured to perform real-time analysis using a machine-learning model. The system further includes a detection module for identifying structural deformation, a localization module for associating detected anomalies with geographic coordinates and timestamps, and an output module for generating annotated thermal images and geotagged data. The integrated system enables real-time detection and mapping of pipeline deformation to support predictive maintenance. (See Fig.1)

FORM 2
THE PATENT ACT 1970
(39 OF 1970)
&
THE PATENT RULES, 2003
COMPLETE SPECIFICATION
(SEE SECTION 10)

1. TITLE OF THE INVENTION

**UAV-BASED SYSTEM FOR DETECTION AND GEOLOCATION OF
UNDERGROUND PIPELINE STRUCTURAL DEFORMATION**

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3. PREAMBLE TO THE DESCRIPTION

**THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES
THE INVENTION AND THE MANNER IN WHICH IT IS TO BE
PERFORMED.**

4. DESCRIPTION

UAV-BASED SYSTEM FOR DETECTION AND GEOLOCATION OF UNDERGROUND PIPELINE STRUCTURAL DEFORMATION

TECHNICAL FIELD

[0001] The present invention relates generally to unmanned aerial vehicle (UAV)-based inspection systems. More particularly, the invention relates to a system and method for detecting structural deformation in underground pipelines using thermal imaging, onboard edge computing, and machine-learning-based image analysis. The invention further relates to geolocation and mapping of detected deformation events for infrastructure monitoring and predictive maintenance.

BACKGROUND OF THE INVENTION

[0002] Underground pipelines are widely used for transporting fluids such as oil, gas, and water across extended distances. Over time, such pipelines are subject to structural degradation including cracks, subsidence, bending, and stress-induced deformation caused by corrosion, soil movement, mechanical stress, and environmental conditions.

[0003] Conventional inspection approaches include manual surveys, ground-based sensing techniques, and aerial inspection using unmanned aerial vehicles. However, these approaches are limited in their ability to detect subsurface structural anomalies, particularly at early stages prior to leakage or catastrophic failure.

[0004] Certain existing systems employ thermal imaging or machine-learning-based analysis; however, such systems are typically directed toward leak detection or surface-level anomaly identification rather than structural deformation of underground pipelines.

[0005] Further, many existing inspection systems rely on cloud-based or ground-station processing architectures, which introduce latency, increase dependency on communication infrastructure, and limit real-time decision-making in field environments.

[0006] Accordingly, there exists a need for an integrated system capable of detecting structural deformation in underground pipelines using thermal sensing, performing real-time analysis onboard an aerial platform, and associating detected anomalies with precise geolocation data.

SUMMARY OF THE INVENTION

[0007] An object of the present invention is to provide a UAV-based system capable of detecting structural deformation in underground pipelines using thermal imaging.

[0008] Another object of the invention is to enable real-time analysis of thermal data using a machine-learning model executed on an onboard edge computing device.

[0009] A further object of the invention is to detect and classify deformation patterns including cracks, subsidence, and bending based on thermal anomaly signatures.

[0010] Another object of the invention is to associate detected deformation events with geographic coordinates and timestamps to facilitate mapping and maintenance planning.

[0011] A further object of the invention is to reduce reliance on remote processing infrastructure by performing inference locally on the UAV platform.

[0012] In an embodiment, the invention comprises a UAV platform carrying a thermal imaging module configured to capture a radiometric temperature matrix corresponding to subsurface thermal variations, a preprocessing module configured to convert the temperature matrix into a color-mapped thermal image and extract regions of interest, an onboard edge computing module configured to process the thermal image in real time, a machine-learning-based detection module configured to detect structural deformation, a localization module configured to associate detected deformation with geographic coordinates and timestamps, and an output module configured to generate annotated thermal images and geotagged data.

[0013] In an embodiment, the thermal imaging module captures a radiometric temperature matrix which is transformed into a visual representation through color mapping prior to machine-learning-based analysis.

[0014] In another embodiment, the edge computing module is configured to execute the detection module using a pre-trained model stored locally on the UAV platform.

[0015] The integrated operation of sensing, processing, detection, and geolocation enables early identification of structural deformation in underground pipelines and supports predictive maintenance.

[0016] The invention provides a technical effect of enabling real-time detection of subsurface structural deformation using thermal data transformation and onboard processing, thereby reducing latency and improving detection accuracy compared to remote processing system.

[0017] In an embodiment, subsurface structural deformation produces spatially diffused thermal gradients characterized by non-linear heat dispersion patterns, which differ from localized high-intensity thermal anomalies associated with fluid leakage or surface-level defects. The system is configured to analyze spatial continuity, gradient distribution, and thermal diffusion characteristics of the temperature matrix to distinguish deformation-induced anomalies from other environmental or operational thermal variations.

BRIEF DESCRIPTION OF THE DRAWINGS

[0018] FIG. 1 illustrates a workflow of a UAV-based system for detecting structural deformation in underground pipelines, including thermal data acquisition, preprocessing, onboard processing, deformation detection, geolocation, and output generation.

DETAILED DESCRIPTION

[0019] Referring to FIG. 1, the invention provides a UAV-based system configured to detect structural deformation in underground pipelines through analysis of subsurface thermal variations. The system is implemented on a UAV platform (100) that traverses a pipeline inspection region while carrying sensing, processing, and localization components, enabling real-time acquisition and analysis of thermal data corresponding to subsurface structural conditions.

[0020] During operation, a thermal imaging module (102) mounted on the UAV platform (100) captures radiometric thermal data in the form of a temperature matrix representing spatial temperature distributions across the inspection region. The captured temperature matrix reflects subsurface thermal behavior associated with the underground pipeline, wherein structural deformation such as bending, subsidence, or stress-induced distortion produces spatially distributed thermal gradients rather than localized high-intensity anomalies typically associated with leakage or surface defects.

[0021] The captured thermal data is processed to generate a visual representation suitable for computational analysis. In this regard, the temperature matrix is transformed into a color-mapped thermal image using a predefined color mapping scheme that enhances visibility of thermal gradients and spatial diffusion patterns. The processing further includes identification of regions corresponding to the pipeline

path, thereby isolating areas of interest within the thermal image for subsequent analysis.

[0022] The processed thermal image is analyzed in real time using an onboard edge computing module (106) disposed on the UAV platform (100), thereby eliminating dependence on remote processing infrastructure and enabling immediate interpretation of thermal data during flight. A detection module (108), operatively associated with the edge computing module (106), applies a machine-learning-based model to the processed thermal image to identify structural deformation patterns. The detection is based on analysis of spatial continuity, gradient distribution, and diffusion characteristics of thermal signatures, allowing differentiation between deformation-induced anomalies and other environmental or operational thermal variations.

[0023] When deformation is detected, the system associates the detected event with spatial and temporal information. A localization module (110) determines geographic coordinates corresponding to the UAV position at the time of detection and assigns a timestamp to each detected anomaly, thereby enabling precise geolocation of structural deformation events along the pipeline.

[0024] The system further generates output data representing detected deformation. An output module (112) produces annotated thermal images in which deformation regions are identified and marked, along with associated metadata including location coordinates and detection confidence. The output data may be used for visualization,

mapping, and maintenance planning, enabling operators to identify critical regions requiring intervention.

[0025] The operation of the system thus represents an integrated workflow in which thermal data acquisition, transformation, real-time processing, deformation detection, geolocation, and output generation are performed onboard the UAV platform (100). By performing inference locally and analyzing thermal gradient patterns associated with subsurface deformation, the system enables early identification of structural anomalies prior to leakage or failure, thereby supporting predictive maintenance and reducing operational risks.

[0026] In an embodiment, the detection module utilizes a pre-trained machine-learning model stored locally on the edge computing module (106), enabling execution of inference during UAV operation without external connectivity. In another embodiment, the system distinguishes structural deformation from leakage or surface-level anomalies by analyzing non-linear heat dispersion patterns and spatial continuity of thermal gradients, thereby improving detection accuracy in complex environmental conditions.

5. CLAIMS

I/We Claim:

1. A UAV-based system for detecting structural deformation in underground pipelines, the system comprising:
 - a UAV platform configured to traverse a pipeline inspection region;
 - a thermal imaging module configured to capture a temperature matrix representing

- subsurface thermal variations;
 - a preprocessing module configured to convert the temperature matrix into a color-mapped thermal image and to extract one or more regions of interest corresponding to a pipeline zone;
 - an onboard edge computing module configured to process the thermal image in real time without reliance on remote processing infrastructure;
 - a machine-learning-based detection module configured to process the color-mapped thermal image to detect structural deformation based on spatial thermal gradient patterns corresponding to subsurface heat diffusion;
 - a localization module configured to associate detected deformation with geographic coordinates and timestamps; and
 - an output module configured to generate annotated thermal images and geotagged deformation data.
2. A method for detecting structural deformation in underground pipelines using a UAV-based system, the method comprising:
 - capturing a temperature matrix representing subsurface thermal variations using a thermal imaging module mounted on a UAV;
 - converting the temperature matrix into a color-mapped thermal image;
 - preprocessing the thermal image to extract one or more regions of interest corresponding to a pipeline region;
 - performing real-time inference on the thermal image using a machine-learning model deployed on an onboard edge computing module;
 - detecting structural deformation based on thermal anomaly patterns;
 - associating detected deformation with geographic coordinates and timestamps; and
 - generating geotagged outputs including annotated thermal images.
 3. The system as claimed in claim 1, wherein the color-mapped thermal image is generated using a plasma-based color mapping scheme configured to enhance visibility of thermal gradients.
 4. The system as claimed in claim 1, wherein the preprocessing module is further configured to perform calibration and noise reduction on the captured temperature matrix prior to image conversion.
 5. The system as claimed in claim 1, wherein the machine-learning-based detection module comprises a convolutional neural network trained to identify deformation patterns including cracks, subsidence, and bending.
 6. The system as claimed in claim 1, wherein the onboard edge computing module comprises a single-board computing device configured to execute inference locally on the UAV platform.
 7. The system as claimed in claim 1, wherein the detection module is configured to generate bounding boxes and associated confidence scores corresponding to detected deformation regions.
 8. The system as claimed in claim 1, wherein the localization module utilizes a satellite-based positioning system to obtain geographic coordinates of detected deformation.
 9. The system as claimed in claim 1, wherein the output module is configured to provide geotagged visualization of detected deformation on a monitoring interface.

10. The method as claimed in claim 2, wherein the step of detecting structural deformation comprises classifying thermal anomaly patterns using a trained convolutional neural network model.

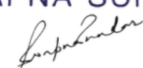
6. SIGNATURE WITH DATE

SWAPNA SUNDAR

PATENT AGENT (IN-PA837)

7. ABSTRACT

A UAV-based system for detecting structural deformation in underground pipelines is disclosed. The system comprises a thermal imaging module configured to capture subsurface thermal variations, a preprocessing module configured to convert thermal data into color-mapped images, and an onboard edge computing module configured to perform real-time analysis using a machine-learning model. The system further includes a detection module for identifying structural deformation, a localization module for associating detected anomalies with geographic coordinates and timestamps, and an output module for generating annotated thermal images and geotagged data. The integrated system enables real-time detection and mapping of pipeline deformation to support predictive maintenance. (See Fig.1)

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